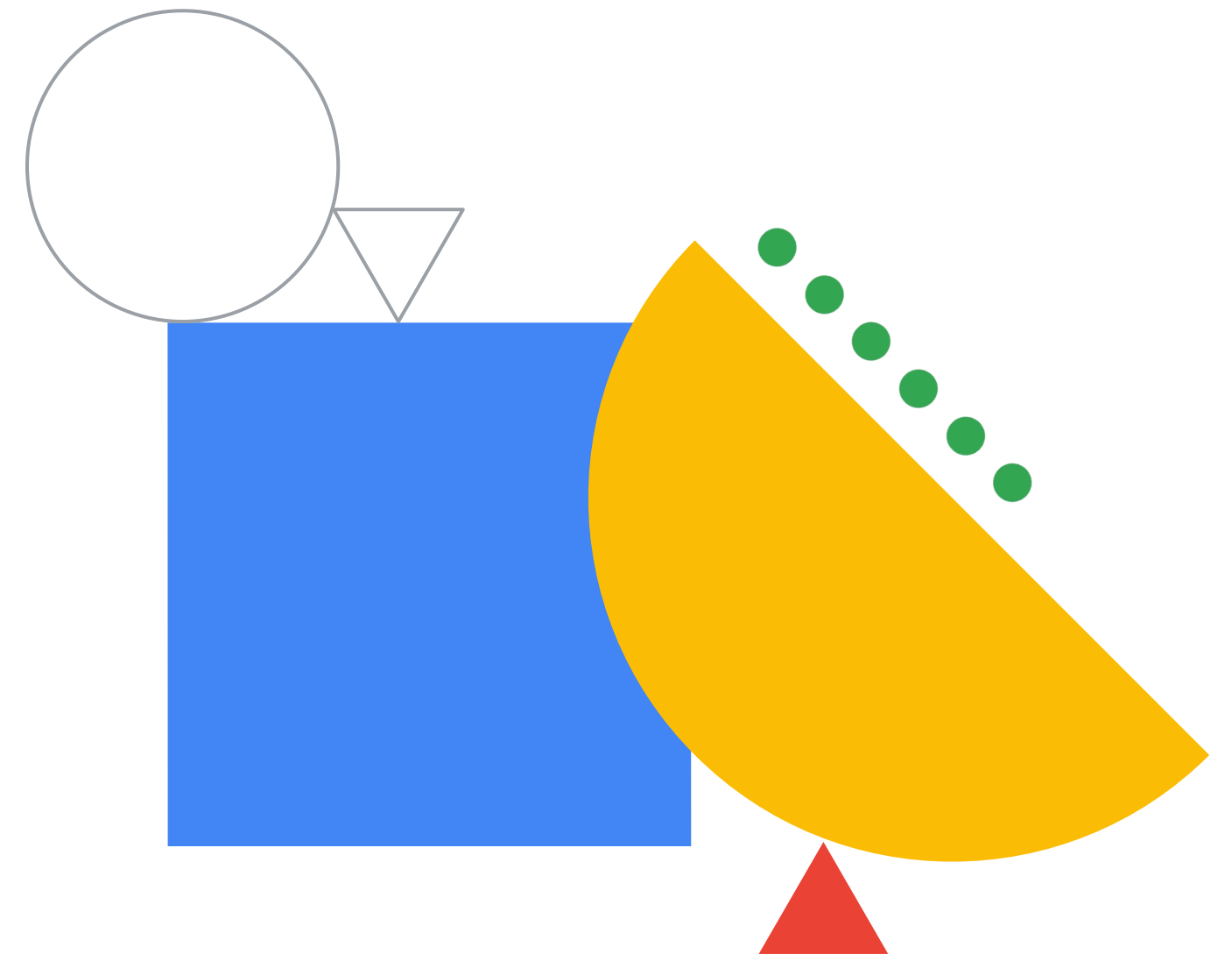


Preparing for Your Associate Cloud Engineer Journey

Module 2: Planning and Configuring Cloud Solutions



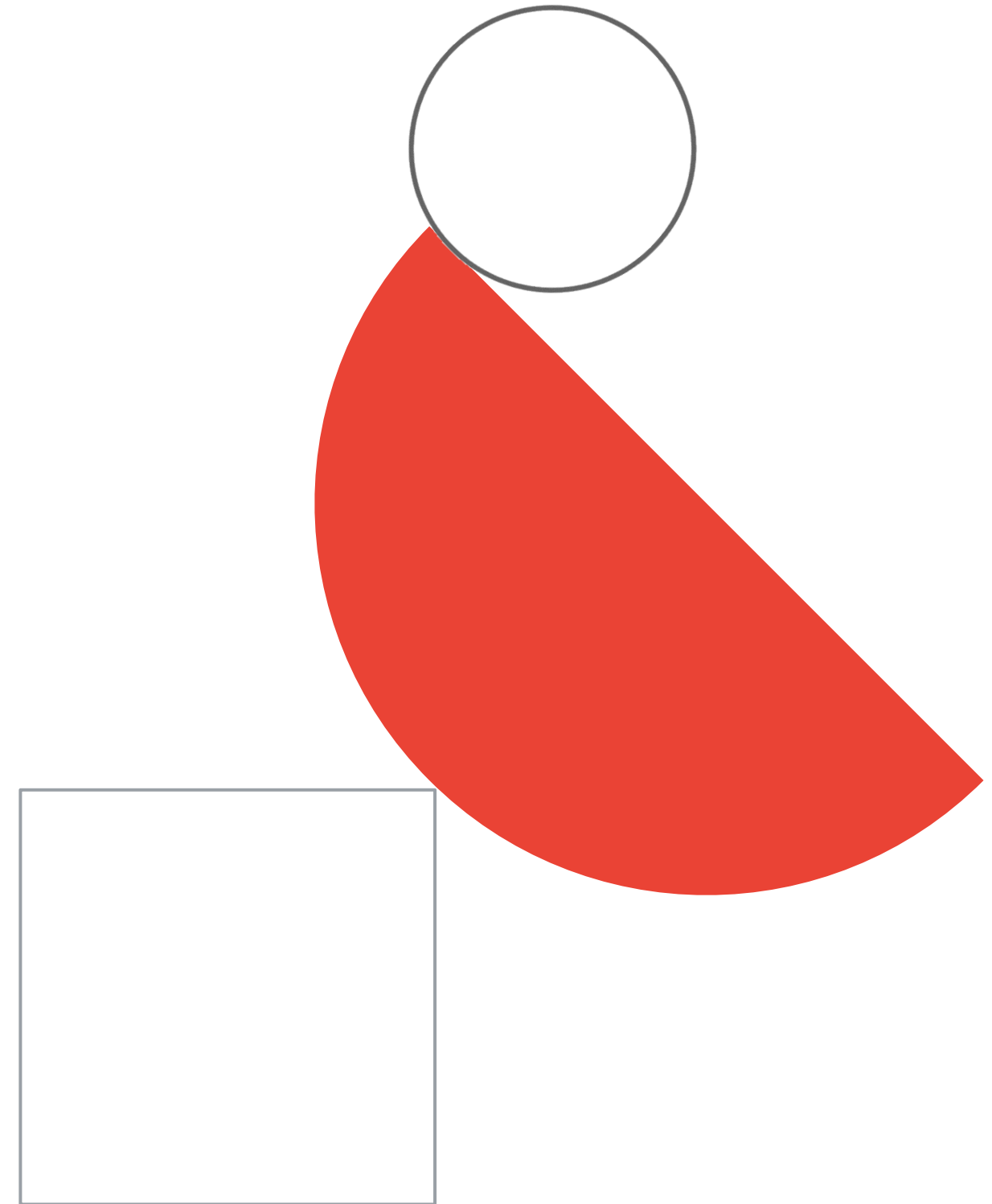


Module agenda

- 01 Selecting resources for Cymbal Superstore's cloud solutions
- 02 Diagnostic questions
- 03 Review and study planning

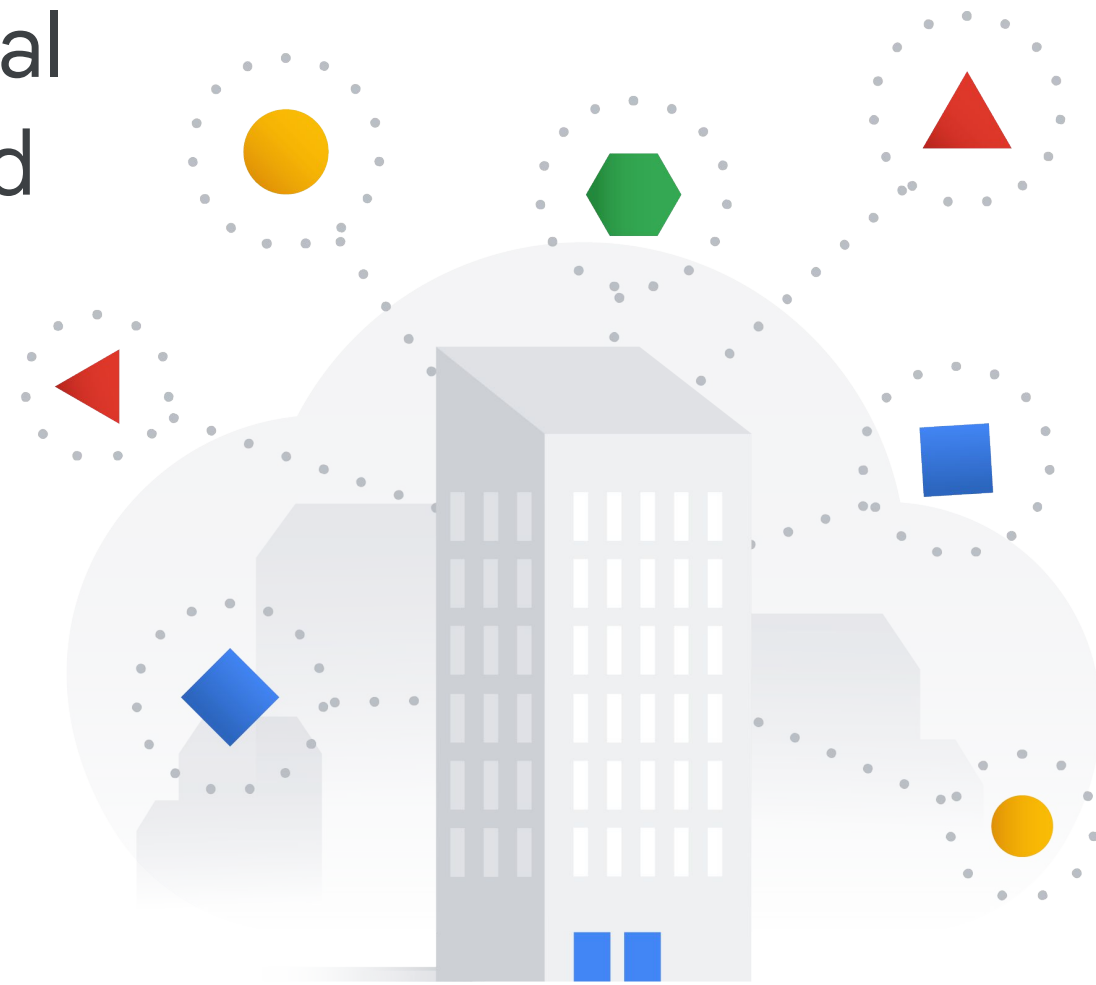


Selecting resources for Cymbal Superstore's cloud solutions



The next step:

planning and
configuring Cymbal
Superstore's cloud
solutions



- Planning and estimating Google Cloud pricing using the Pricing Calculator
- Planning and configuring compute resources
- Planning and configuring data storage options
- Planning and configuring network resources



Cymbal Superstore's existing applications



Ecommerce

Cymbal Superstore has an existing web application that provides an interface for customers to look at and order products.

Requirements:

- Compute: Container architecture
- Data: Relational backend
- Networking: Needs to be globally available
- Need analytical capabilities to inform marketing efforts



Transportation Management

Delivery services is becoming an important aspect of Cymbal Superstore's customer interactions. Cymbal Superstore would like to use Google Services to keep track of truck location.

Requirements:

- Dashboard of truck location in near real-time
- Analysis of truck mileage for preventive maintenance



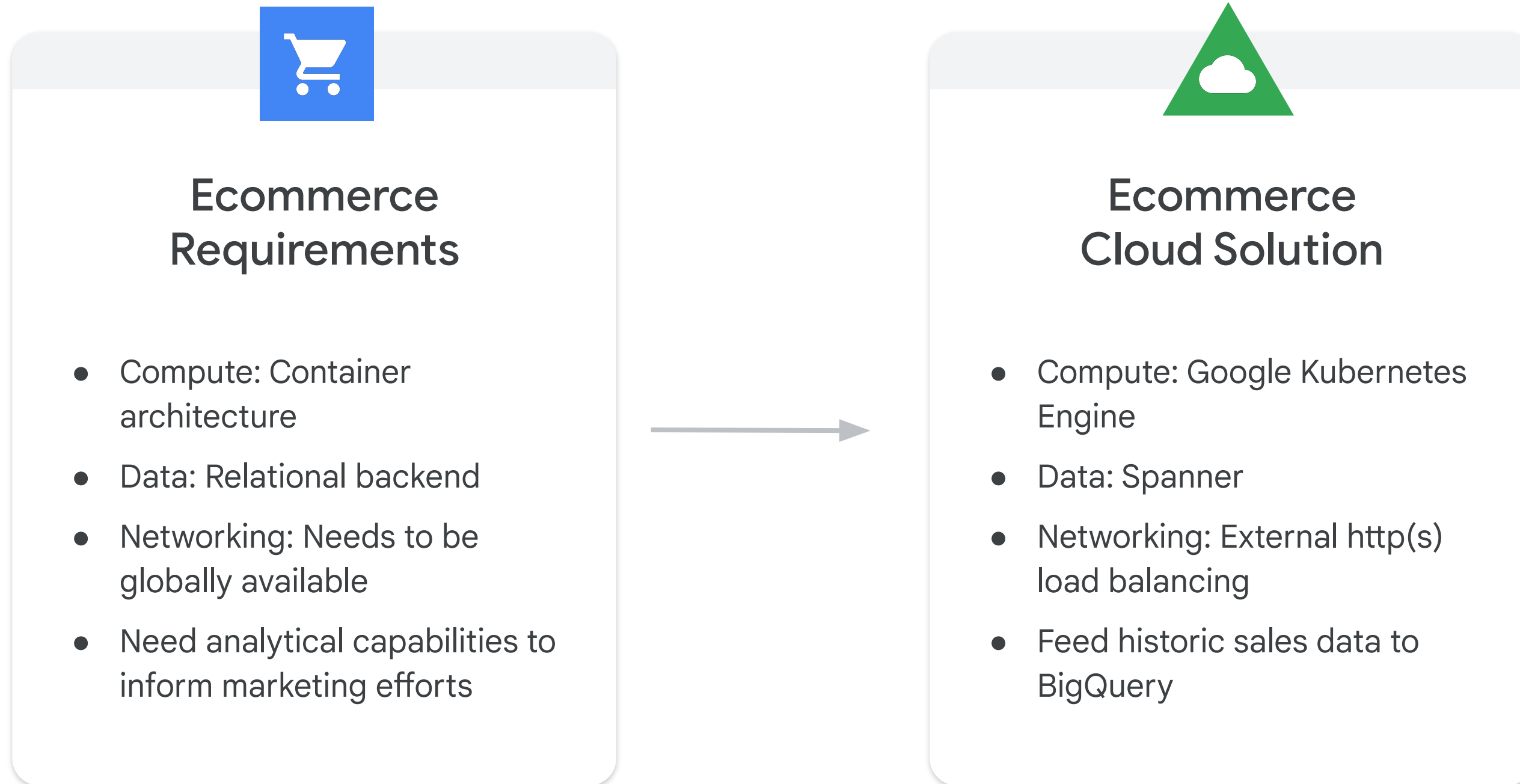
Supply Chain

Cymbal Superstore has decided to migrate their legacy supply chain application to the cloud

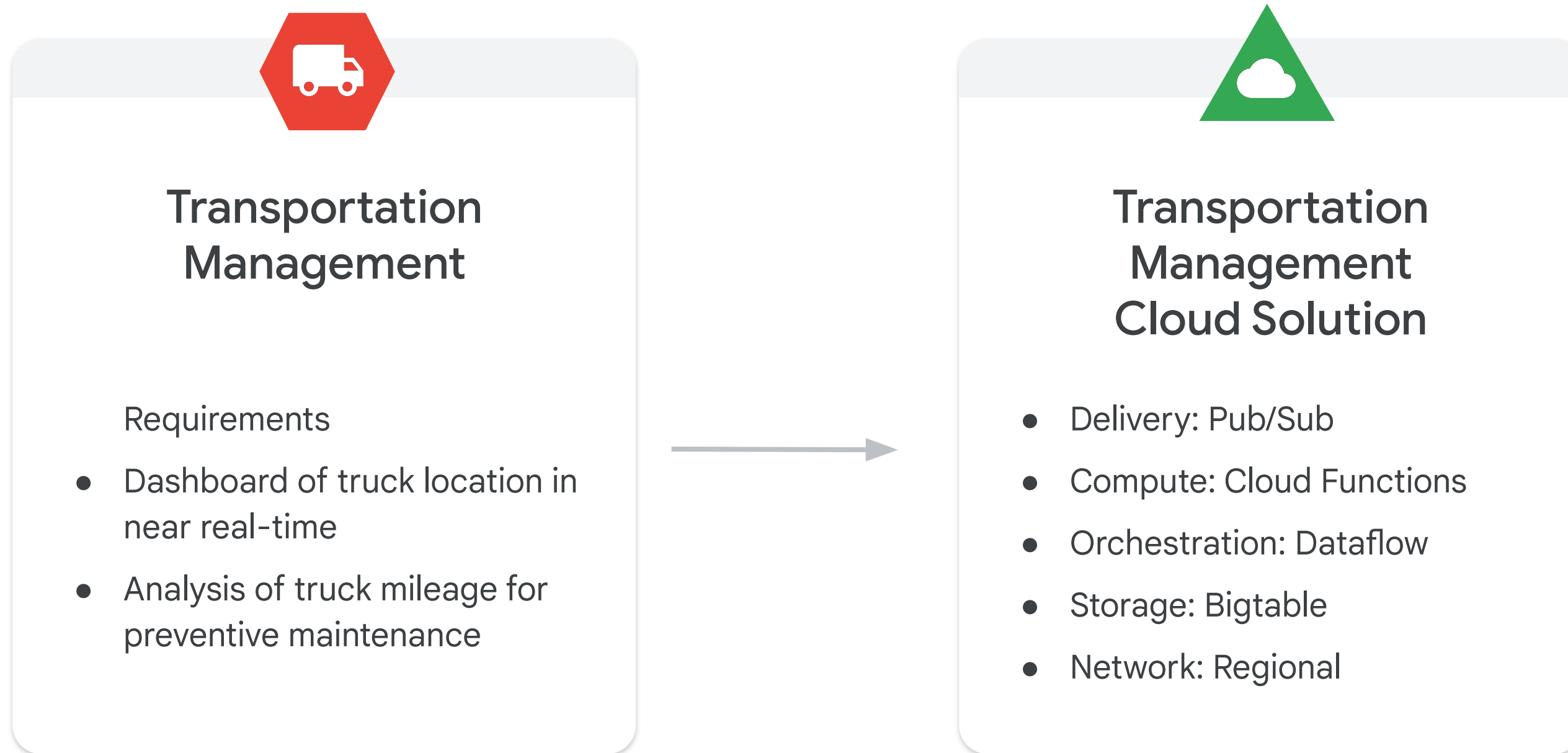
Requirements:

- Available local to their HQ
- Currently implemented in virtual machines with a LINUX operating system and a LAMP stack

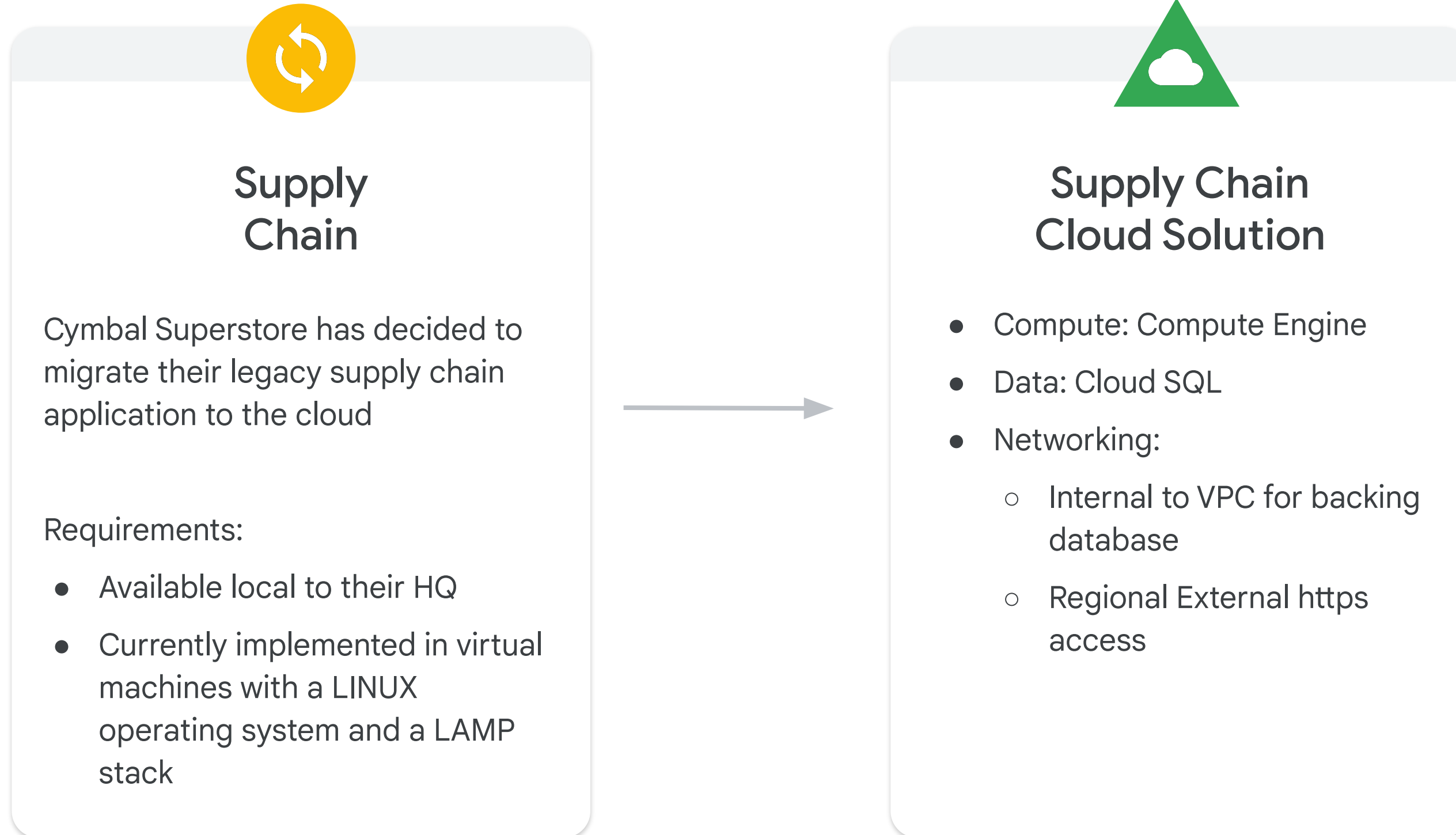
Cymbal Superstore's ecommerce solution



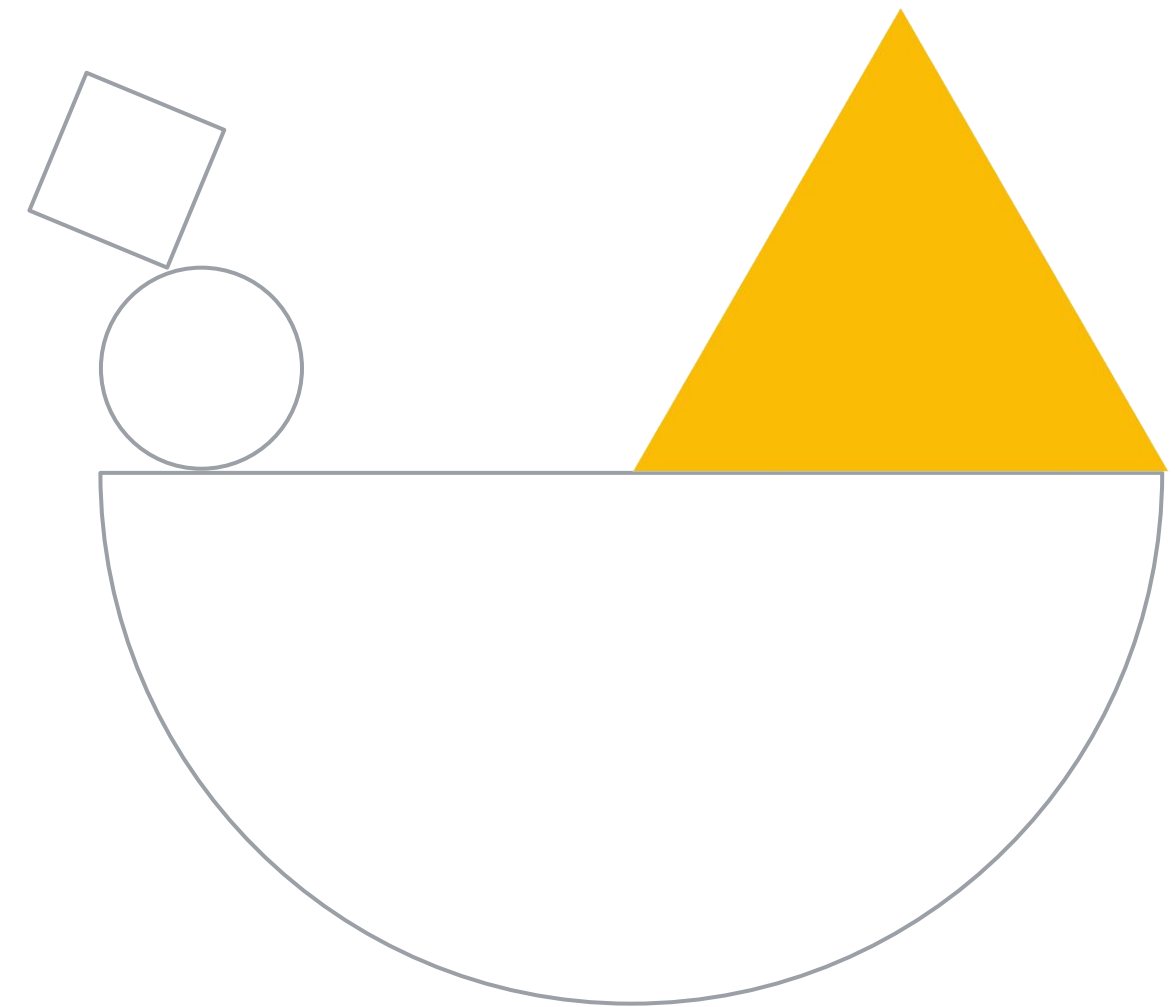
Cymbal Superstore's transportation management solution



Cymbal Superstore's supply chain solution



Diagnostic questions

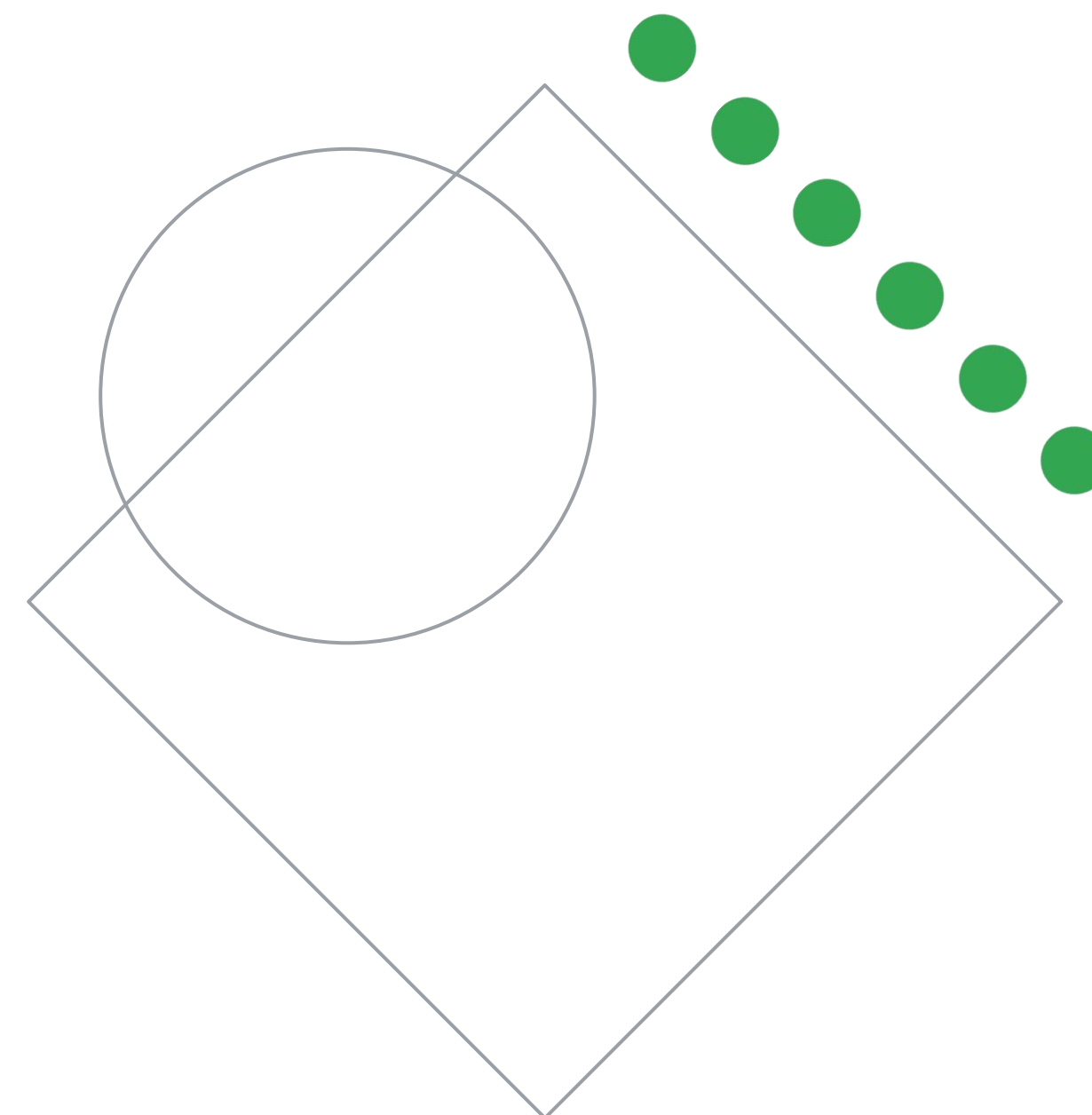


Please complete the diagnostic questions now

- The diagnostic questions are available in the workbook.

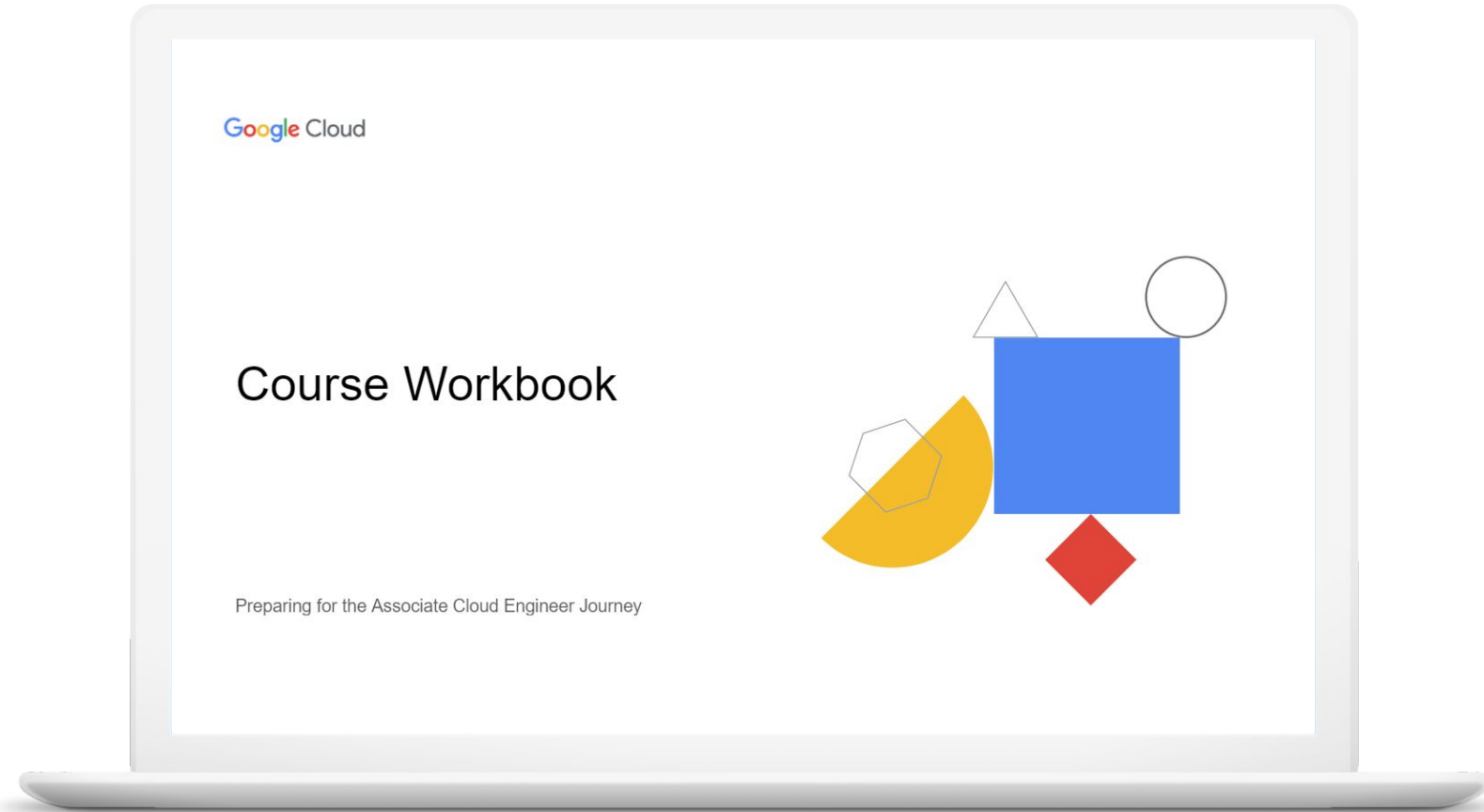


Review and study planning



Your study plan:

Planning and configuring cloud solutions



2.1

Planning and estimating using the Pricing Calculator

2.2

Planning and configuring compute resources

2.3

Planning and configuring data storage options

2.4

Planning and configuring network resources

2.1 | Planning and estimating using the Pricing Calculator

2.1 | Diagnostic Question 01 Discussion

The projected amount of cloud storage required for Cymbal Superstore to enable users to post pictures for project reviews is 10 TB of immediate access storage in the US and 30 TB of storage for historical posts in a bucket located near Cymbal Superstore's headquarters. The contents of this bucket will need to be accessed once every 30 days. You want to estimate the cost of these storage resources to ensure this is economically feasible.

What should you do?

- A. Use the pricing calculator to estimate the costs for 10 TB of regional Standard storage, 30 TB of regional Coldline storage, and egress charges for reads from storage.
- B. Use the pricing calculator to estimate the price for 10 TB of regional Standard storage, 30 TB of regional Nearline storage, and ingress charges for posts to the bucket.
- C. Use the pricing calculator to estimate the price for 10 TB of multi-region standard storage, 30 TB for regional Coldline storage, and ingress charges for posts to the bucket.
- D. Use the pricing calculator to estimate the price for 10 TB of multi-region Standard storage, 30 TB for regional Nearline, and egress charges for reads from the bucket.



2.1 | Diagnostic Question 01 Discussion

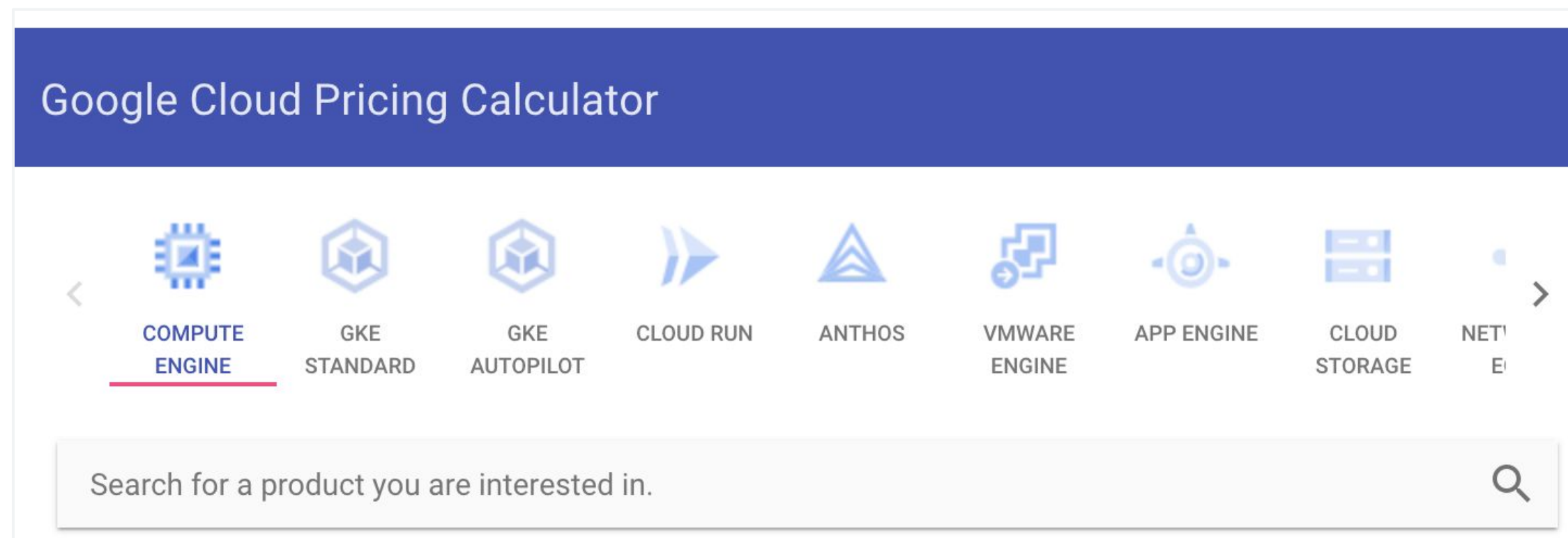
The projected amount of cloud storage required for Cymbal Superstore to enable users to post pictures for project reviews is 10 TB of immediate access storage in the US and 30 TB of storage for historical posts in a bucket located near Cymbal Superstore's headquarters. The contents of this bucket will need to be accessed once every 30 days. You want to estimate the cost of these storage resources to ensure this is economically feasible.

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- C. Use the pricing calculator to estimate the price for 10 TB of multi-region standard storage, 30 TB for regional Coldline storage, and ingress charges for posts to the bucket.
- D. Use the pricing calculator to estimate the price for 10 TB of multi-region Standard storage, 30 TB for regional Nearline, and egress charges for reads from the bucket.**



Pricing Calculator



<https://cloud.google.com/products/calculator/>

1

Select a product from scrolling list at top of the form

2

Form for each product will show expense variables

3

Enter target configuration in form

4

Submit each section to add to your overall estimate

Pricing Calculator

Total Estimated Cost: USD 4,958.48 per 1 month

Estimate Currency

USD - US Dollars ▼

Adjust Estimate Timeframe

1 day

1 week

1 month

1 quarter

1 year

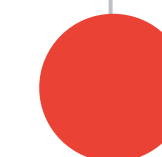
3 years

EMAIL ESTIMATE

SAVE ESTIMATE



Total estimated cost:
daily, weekly, monthly,
quarterly, yearly and
3-year increments



Cost is only an estimate



Question to ask:
How closely your
estimated usage
matches your actual
usage?



Not a binding contract,
just a planning tool

2.1

Planning and estimating
using the Pricing Calculator

Courses

[Architecting with Google Compute Engine](#)

- M3 Virtual Machines
- M6 Resource Management



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[Essential Google Cloud Infrastructure: Foundation](#)

- M3 Virtual Machines

[Essential Google Cloud Infrastructure: Core Services](#)

- M3 Resource Management



Documentation

[Google Cloud Pricing Calculator](#)

2.2 | Planning and configuring compute resources

Considerations include:

- Selecting appropriate compute choices for a given workload
(e.g., Compute Engine, Google Kubernetes Engine, Cloud Run, Cloud Functions)
- Using preemptible VMs and custom machine types as appropriate

2.2 | Diagnostic Question 02 Discussion



Cymbal Superstore decides to migrate their supply chain application to Google Cloud. You need to configure specific operating system dependencies.

What should you do?

- A. Implement an application using containers on Cloud Run.
- B. Implement an application using code on App Engine.
- C. Implement an application using containers on Google Kubernetes Engine.
- D. Implement an application using virtual machines on Compute Engine.

2.2 | Diagnostic Question 02 Discussion



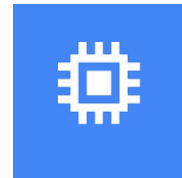
Cymbal Superstore decides to migrate their supply chain application to Google Cloud. You need to configure specific operating system dependencies.

What should you do?

- A. Implement an application using containers on Cloud Run.
- B. Implement an application using code on App Engine.
- C. Implement an application using containers on Google Kubernetes Engine.
- D. Implement an application using virtual machines on Compute Engine.**



Infrastructure as a service:



Google Compute Engine

Virtual machines running in
Google's global data centers

Use When You
Need...

- Complete control
- Ability to make OS level changes
- To be able to move to the cloud without rewriting your code
- To use custom VM images

Typical Use Cases

- Any workload requiring a specific OS or configuration
- On-premises software that you want to run in the cloud



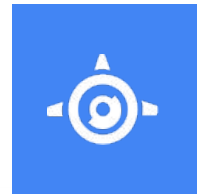
Google Kubernetes Engine

Logical infrastructure powered by [Kubernetes](#),
the open source container orchestration system

- No dependencies on a specific OS
- Increased velocity and operability
- To manage containers in production

- Containerized workloads
- Cloud-native distributed systems
- Hybrid applications

Platform as a service:



Google App Engine

Flexible, zero-ops platform
for building apps

- To just focus on writing code
- Developer velocity
- To minimize operational overhead

- Web sites
- Apps (of course!)
- Gaming back ends
- IoT applications

Use When You
Need...

Typical Use Cases

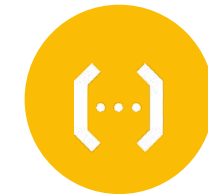


Google Cloud Run

Deploy code or containers that
listens for requests or events

- Scales to meet demand
- Pay for what you use
- Supports API endpoints

- Web frameworks
- Microservices



Google Cloud Functions

Serverless execution environment for
building and connecting cloud services

- For event-driven workloads
- Scales to meet demand
- Minimal configuration

- Statistical analysis
- Image thumbnail generation
- Post a comment to a Slack channel
after a GitHub commit

2.2 | Diagnostic Question 03 Discussion



Cymbal Superstore decides to pilot a cloud application for their point of sale system in their flagship store. You want to focus on code and develop your solution quickly, and you want your code to be portable.

How do you proceed?

- A. SSH into a Compute Engine VM and execute your code.
- B. Package your code to a container image and post it to Cloud Run.
- C. Implement a deployment manifest and run `kubectl apply` on it in Google Kubernetes Engine.
- D. Code your solution in Cloud Functions.

2.2 | Diagnostic Question 03 Discussion



Cymbal Superstore decides to pilot a cloud application for their point of sale system in their flagship store. You want to focus on code and develop your solution quickly, and you want your code to be portable.

How do you proceed?

- A. SSH into a Compute Engine VM and execute your code.
- B. Package your code to a container image and post it to Cloud Run.**
- C. Implement a deployment manifest and run `kubectl apply` on it in Google Kubernetes Engine.
- D. Code your solution in Cloud Functions.



2.2 | Diagnostic Question 04 Discussion



An application running on a highly-customized version of Ubuntu needs to be migrated to Google Cloud. You need to do this in the least amount of time with minimal code changes.

How should you proceed?

- A. Create Compute Engine Virtual Machines and migrate the app to that infrastructure.
- B. Deploy the existing application to App Engine.
- C. Deploy your application in a container image to Cloud Run.
- D. Implement a Kubernetes cluster and create pods to enable your app.

2.2 | Diagnostic Question 04 Discussion



An application running on a highly-customized version of Ubuntu needs to be migrated to Google Cloud. You need to do this in the least amount of time with minimal code changes.

How should you proceed?

- A. **Create Compute Engine Virtual Machines and migrate the app to that infrastructure.**
- B. Deploy the existing application to App Engine.
- C. Deploy your application in a container image to Cloud Run.
- D. Implement a Kubernetes cluster and create pods to enable your app.



2.2 | Diagnostic Question 05 Discussion



You want to deploy a microservices application. You need full control of how you manage containers, reliability, and autoscaling, but don't want or need to manage the control plane.

- A. Cloud Run
- B. App Engine
- C. Google Kubernetes Engine
- D. Compute Engine

Which compute option should you use?

2.2 | Diagnostic Question 05 Discussion



You want to deploy a microservices application. You need full control of how you manage containers, reliability, and autoscaling, but don't want or need to manage the control plane.

- A. Cloud Run
- B. App Engine
- C. Google Kubernetes Engine**
- D. Compute Engine



Which compute option should you use?

2.2 | Planning and configuring compute resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M3 Virtual Machines and Networks in the Cloud
- M5 Containers in the Cloud
- M6 Applications in the Cloud

[Getting Started with Google Kubernetes Engine](#)

- M2 Introduction to Containers and Kubernetes

[Architecting with Google Compute Engine](#)

- M3 Virtual Machines



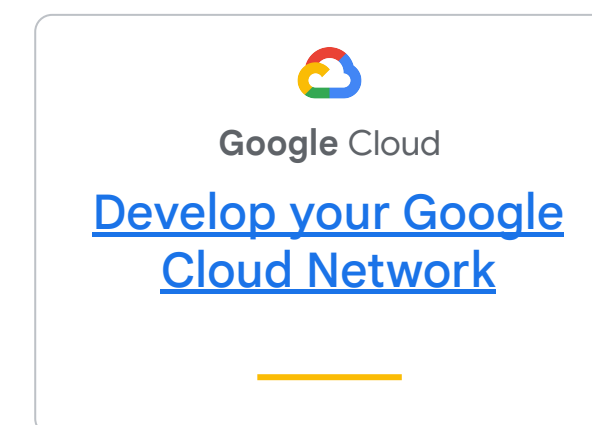
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[Essential Google Cloud Infrastructure: Foundation](#)

- M3 Virtual Machines



Skill Badge



Documentation

[Choosing the right compute option in GCP: a decision tree](#)

[Application Hosting Options](#)

[Tutorials | Compute Engine Documentation](#)

2.3 | Planning and configuring data storage options

Considerations include:

- Product choice
(e.g., Cloud SQL, BigQuery, Firestore, Spanner, Bigtable)
- Choosing storage options
(e.g., Zonal persistent disk, Regional balanced persistent disk, standard, Nearline, Coldline, Archive)

2.3 | Diagnostic Question 06 Discussion

Cymbal Superstore needs to analyze whether they met quarterly sales projections. Analysts assigned to run this query are familiar with SQL.

- A. BigQuery
- B. Cloud SQL
- C. Spanner
- D. Firestore

What data solution should they implement?



2.3 | Diagnostic Question 06 Discussion



Cymbal Superstore needs to analyze whether they met quarterly sales projections. Analysts assigned to run this query are familiar with SQL.

A. BigQuery

B. Cloud SQL

C. Spanner

D. Firestore



What data solution should they implement?

Comparing Data Storage and Database Options

Relational		Non-relational		Object	Warehouse
					
Cloud SQL	Spanner	Firestore	Bigtable	Cloud Storage	BigQuery
Good for: Web frameworks	Good for: RDBMS+scale, HA, HTAP	Good for: Hierarchical, mobile, web	Good for: Heavy read + write, events	Good for: Binary or object data	Good for: Enterprise data warehouse
Such as: CMS, eCommerce	Such as: User metadata, Ad/Fin/MarTech	Such as: User profiles, Game State	Such as: AdTech, financial, IoT	Such as: Images, media serving, backups	Such as: Analytics, dashboards

2.3 | Diagnostic Question 07 Discussion

Cymbal Superstore's supply chain application frequently analyzes large amounts of data to inform business processes and operational dashboards.

What storage class would make sense for this use case?

- A. Archive
- B. Coldline
- C. Nearline
- D. Standard



2.3 | Diagnostic Question 07 Discussion



Cymbal Superstore's supply chain application frequently analyzes large amounts of data to inform business processes and operational dashboards.

- A. Archive
- B. Coldline
- C. Nearline
- D. Standard

What storage class would make sense for this use case?



Storage Classes and use cases summary

	Standard	Nearline	Coldline	Archive
Use when you need...	• No retrieval cost • No minimum storage duration	• Very low cost per GB stored and can accept higher per-operation costs • 30-day minimum storage duration	• Even lower cost per GB stored and can accept higher per-operation costs • 90-day minimum storage duration	• Lowest cost per GB stored and can accept the highest per-operation costs • 365-day minimum storage duration
Typical use cases	“Hot” data and/or stored for only brief periods of time like data-intensive computations	Infrequently (i.e., no more than once per month) accessed data. Ideal for back-up and serving long-tail multimedia content.	Very infrequently accessed data - ie, once a year. Typically this is for disaster recovery, or for financial data that has to be kept for a certain length of time to meet regulatory needs.	Data archiving, online backup, and disaster recovery

2.3 | Diagnostic Question 08 Discussion



Cymbal Superstore has a need to populate visual dashboards with historical time-based data. This is an analytical use-case.

Which two storage solutions could they use?

- A. BigQuery
- B. Cloud Storage
- C. Firestore
- D. Cloud SQL
- E. Bigtable

2.3 | Diagnostic Question 08 Discussion



Cymbal Superstore has a need to populate visual dashboards with historical time-based data. This is an analytical use-case.

Which two storage solutions could they use?

A. BigQuery



B. Cloud Storage

C. Firestore

D. Cloud SQL

E. Bigtable



Comparing storage options: use cases

	Firestore	Bigtable	Cloud Storage	Cloud SQL	Spanner	BigQuery
Type	NoSQL document	NoSQL wide column	Blobstore	Relational SQL for OLTP	Relational SQL for OLTP	Relational SQL for OLAP
Best for	Storing, syncing, and querying data	"Flat" data, Heavy read/write, events, analytical data	Structured and unstructured binary or object data	Web frameworks, existing applications	Large-scale database applications (> ~2 TB)	Interactive querying, offline analytics
Use cases	Mobile, web, and server development	AdTech, Financial and IoT data	Images, large media files, backups	User credentials, customer orders	Whenever high I/O, global consistency is needed	Data warehousing

2.3

Planning and configuring data storage options

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M4 Storage in the Cloud

[Architecting with Google Compute Engine](#)

- M5 Storage and Database Services



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[Essential Google Cloud Infrastructure: Core Services](#)

- M2 Storage and Database Services



Skill Badge



Google Cloud

[Set Up an App Dev Environment on Google Cloud](#)

Documentation

[Cloud Storage Options](#)

[Storage classes](#)

[Data lifecycle | Cloud Architecture Center](#)

2.4 | Planning and configuring network resources

Considerations include:

- Differentiating load balancing options
- Identifying resource locations in a network for availability
- Configuring Cloud DNS

2.4 | Diagnostic Question 09 Discussion

Cymbal Superstore is piloting an update to its ecommerce app for the flagship store in Minneapolis, Minnesota. The app is implemented as a three-tier web service with traffic originating from the local area and resources dedicated for it in us-central1. You need to configure a secure, low-cost network load-balancing architecture for it.

How do you proceed?

- A. Implement a premium tier pass-through external https load balancer connected to the web tier as the frontend and a regional internal load balancer between the web tier and backend.
- B. Implement a proxied external TCP/UDP network load balancer connected to the web tier as the frontend and a premium network tier ssl load balancer between the web tier and the backend.
- C. Configure a standard tier proxied external https load balancer connected to the web tier as a frontend and a regional internal load balancer between the web tier and the backend.
- D. Configure a proxied SSL load balancer connected to the web tier as the frontend and a standard tier internal TCP/UDP load balancer between the web tier and the backend.



2.4 | Diagnostic Question 09 Discussion

Cymbal Superstore is piloting an update to its ecommerce app for the flagship store in Minneapolis, Minnesota. The app is implemented as a three-tier web service with traffic originating from the local area and resources dedicated for it in us-central1. You need to configure a secure, low-cost network load-balancing architecture for it.

How do you proceed?

- A. Implement a premium tier pass-through external https load balancer connected to the web tier as the frontend and a regional internal load balancer between the web tier and backend.
- B. Implement a proxied external TCP/UDP network load balancer connected to the web tier as the frontend and a premium network tier ssl load balancer between the web tier and the backend.
- C. Configure a standard tier proxied external https load balancer connected to the web tier as a frontend and a regional internal load balancer between the web tier and the backend.**
- D. Configure a proxied SSL load balancer connected to the web tier as the frontend and a standard tier internal TCP/UDP load balancer between the web tier and the backend.



Google VPC offers a suite of load balancing options

Global HTTP(S)	Global SSL Proxy	Global TCP Proxy	Regional	Regional internal
Layer 7 load balancing based on load	Layer 4 load balancing of non-HTTPS SSL traffic based on load	Layer 4 load balancing of non-SSL TCP traffic	Load balancing of any traffic (TCP, UDP)	Load balancing of traffic inside a VPC
Can route different URLs to different backends	Supported on specific port numbers	Supported on specific port numbers	Supported on any port number	Use for the internal tiers of multi-tier applications

2.4 | Diagnostic Question 10 Discussion

What Google Cloud load balancing option runs at Layer 7 of the TCP stack?

- A. Global http(s)
- B. Global SSL Proxy
- C. Global TCP Proxy
- D. Regional Network



2.4 | Diagnostic Question 10 Discussion

What Google Cloud load balancing option runs at Layer 7 of the TCP stack?

- A. Global http(s)
- B. Global SSL Proxy
- C. Global TCP Proxy
- D. Regional Network



Summary of load balancers

Load balancer	Traffic type	Global/ Regional	External/ Internal	External ports for load balancing
HTTP(S)	HTTP or HTTPS	Global IPv4 IPv6	External	HTTP on 80 or 8080; HTTPS on 443
SSL Proxy	TCP with SSL offload			25, 43, 110, 143, 195, 443, 465, 587, 700, 993, 995, 1883, 5222
TCP Proxy	- TCP without SSL offload - Does not preserve client IP addresses			25, 43, 110, 143, 195, 443, 465, 587, 700, 993, 995, 1883, 5222
Network TCP/UDP	- TCP/UDP without SSL offload - Preserves client IP addresses	Regional IPv4	Internal	Any
Internal TCP/UDP	TCP or UDP			Any
Internal HTTP(S)	HTTP or HTTPS			HTTP on 80 or 8080; HTTPS on 443

2.4 | Planning and configuring network resources

Courses

Documentation

[Google Cloud Fundamentals: Core Infrastructure](#)

- M3 Virtual Machines and Networks in the Cloud
- M4 Storage in the Cloud

[Architecting with Google Compute Engine](#)



- M2 Virtual Networks
- M5 Storage and Database Services
- M9 Load Balancing and Autoscaling

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[Essential Google Cloud Infrastructure: Foundation](#)



[Essential Google Cloud Infrastructure: Core Services](#)

- M2 Virtual Networks
- M2 Storage and Database Services

[Elastic Google Cloud Infrastructure: Scaling and Automation](#)

- M2 Load Balancing and Autoscaling

[Cloud Load Balancing overview](#)

[Cloud Load Balancing](#)

Knowledge Check 1

Which storage class is designed for long term storage has a 365 day minimum storage agreement, and a lower storage price as compared to other storage types?

- A. Standard Storage
- B. Coldline Storage
- C. Nearline Storage
- D. Archive storage



Knowledge Check 1

Which storage class is designed for long term storage has a 365 day minimum storage agreement, and a lower storage price as compared to other storage types?

- A. Standard Storage
- B. Coldline Storage
- C. Nearline Storage
- D. Archive storage



Knowledge Check 2

Which serverless option is based on developing and executing small snippets of code?

- A. Cloud Functions
- B. Cloud Run
- C. BigQuery
- D. Dataflow



Knowledge Check 2

Which serverless option is based on developing and executing small snippets of code?

A. Cloud Functions

B. Cloud Run

C. BigQuery

D. Dataflow

